Known-Plaintext Attack on Hill Cipher

38. Write a C program for Hill cipher succumbs to a known plaintext attack if sufficient plaintext–ciphertext pairs are provided. It is even easier to solve the Hill cipher if a chosen plaintext attack can be mounted. Implement in C programming.

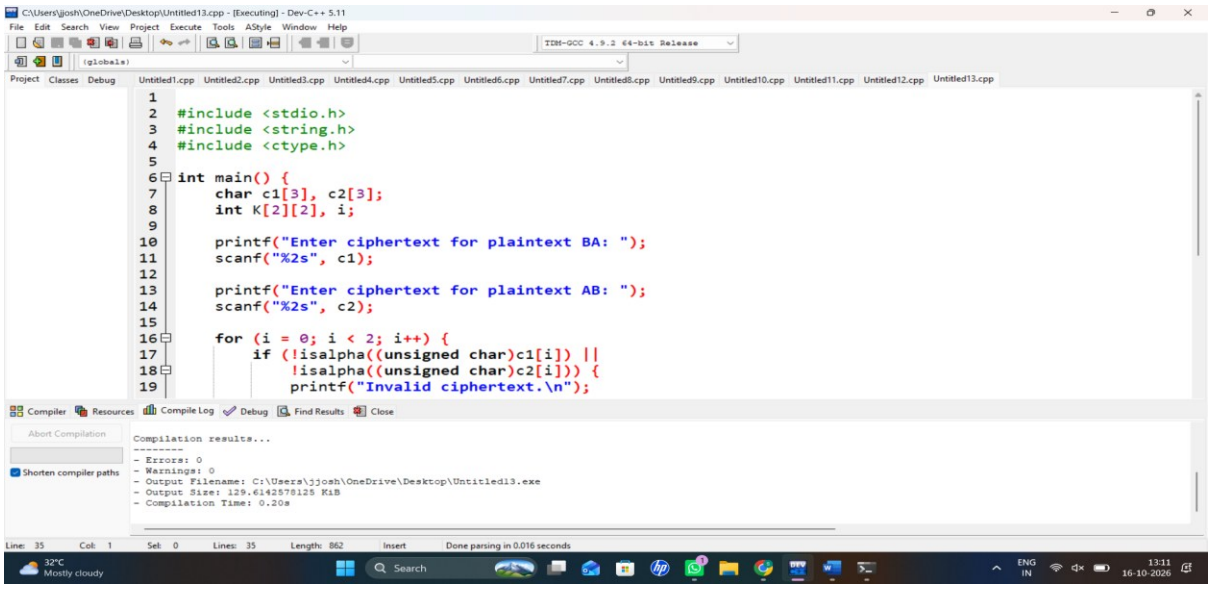
Aim:

To demonstrate a chosen-plaintext attack on a 2 × 2 Hill cipher and recover its encryption key using carefully selected plaintext blocks.

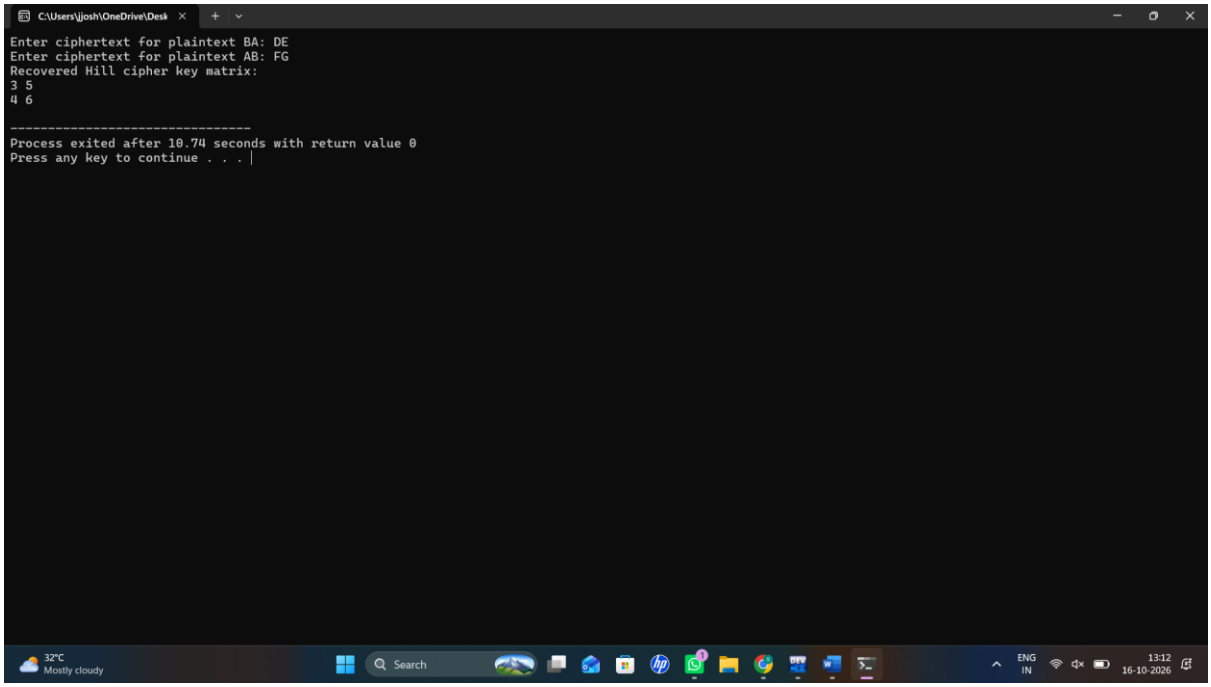
Algorithm:

1. Consider a Hill cipher with a 2 × 2 key matrix.
2. Choose plaintext blocks BA and AB.
3. Convert letters to numbers, with A = 0, B = 1, and so on.
4. Encrypt each block using the Hill cipher.
5. Since these plaintext vectors are \((1,0)\) and \((0,1)\), their ciphertext vectors reveal the two columns of the key matrix.
6. Construct the key matrix from the recovered columns.

C Program:



Input/Output:



Result:

The chosen-plaintext attack recovered the Hill cipher key matrix. Carefully selected plaintext blocks can reveal the columns of the key matrix directly. For a general known-plaintext attack, enough independent plaintext–ciphertext pairs are required, and the plaintext matrix must be invertible modulo 26.